Shubh Vastu AI Astro

Phase 1 System Design

STATUS Proposed

OWNER Shubh Vastu Consultancy

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Authors	Development team
Reviewers	Dr. Jagdish Mohan Mavlankar and client owner
Related docs	UI prototype: ui-prototype.html

Scope	Phase 1 PWA architecture, user experience and operating model

1. Abstract

Launch an Android-installable PWA for Shubh Vastu Consultancy, with Marathi as the default and Hindi and English support. React provides the customer interface; a Firebase backend controls identity, quotas, payments and access to AstrologyAPI.

Phase 1 includes birth charts, AI Guruji, basic Vastu, a gated Prashna workflow and prepaid memberships. Reliable provider data is required for every chart interpretation. The design targets an initial pilot of 100 paying users; this is a planning scenario, not a capacity guarantee.

2. Goals and Non-Goals

Goals	Non-goals
Marathi, Hindi and English customer journeys	Native Android or Play Store release
Verified birth charts and traceable AI interpretations	Guaranteed predictions or invented positions
Paid plans with server-enforced usage limits	Paid PDF reports and marriage matching
Mobile usability, ownership and cost controls	Floor-plan image analysis and automatic renewal

3. Background and Problem Statement

Customers need accessible regional-language guidance and a clear route to Dr. Jagdish Mohan Mavlankar. The product must distinguish traditional interpretations from practical advice and make subscription limits visible before customers pay.

The client owns the source code, domain and service accounts. The PWA cannot call paid providers directly. The backend is the sole authority for paid entitlements and calculation provenance. A full paid MVP exceeds the original INR 10,000 budget; reduce scope if that ceiling remains fixed.[1]

1 Use footnotes for definitions, assumptions, or terms that need precision without interrupting the main narrative.

4. Proposed Architecture

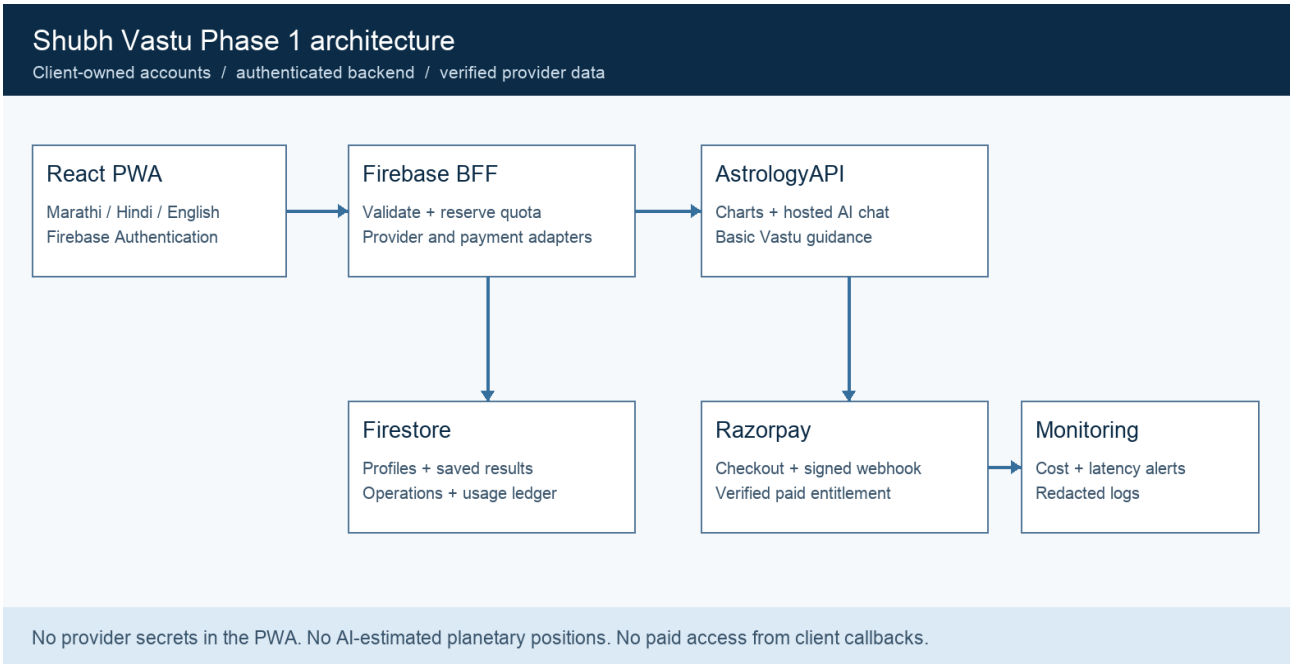


Figure 1. Customer requests cross the backend before reaching paid services.

Core components

Component	Responsibility	Primary storage	Failure behavior
React PWA	Forms, chat, plan display and language selection	Local UI preferences	Offline notice; no new paid calls
Firebase BFF	Validate, reserve quotas and orchestrate providers	Firestore operations	Fail closed; reconcile pending work
AstrologyAPI	Chart JSON, hosted chat and basic Vastu	Vendor service	Show unavailable; never invent facts
Firebase data	User records, saved output and usage ledger	Firestore	Do not start calls if reservation fails
Payments and ops	Razorpay webhooks, alerts and support	Payment ledger and logs	Pending access until verified

5. Request Lifecycle

a. A signed-in customer submits a birth profile, question or Vastu assessment with a unique request ID. The interface shows the relevant allowance before submission.

b. The backend verifies the Firebase ID token, resource ownership, payload limits and rate limits. It normalizes birth time, location, timezone and calculation settings.

c. Load the server-owned plan version, active period, quota counters and consent state. Reuse valid cached natal data only when the complete calculation key matches.

d. Atomically reserve the required allowance and create a pending operation. Call the selected provider adapter outside the Firestore transaction; never make external calls inside a retried transaction.

e. Persist provider responses and provenance before marking the operation complete. AI receives verified chart facts; missing facts produce a clarification or unavailable state, not guessed positions.

f. Use explicit provider deadlines. Retry only safe reads with bounded backoff. A timed-out chat or payment call is uncertain; reconcile it rather than blindly issuing duplicate billable work.

g. Commit usage once on successful delivery; release the reservation on confirmed failure. Return the saved response and remaining quota. Log request IDs, latency and costs without birth details or message bodies.

6. API and Data Contracts

Primary data contract

Field	Type	Required	Description
requestId	UUID	Yes	User-scoped idempotency key; reject changed payload.
operation	Enum	Yes	chart, chat, vastu or prashna; validated allowlist.
profileId	String	Conditional	Owned saved profile for chart-dependent requests.
locale	Enum	Yes	mr, hi or en; Marathi is the default.
inputs	Object	Yes	Validated birth, question or property fields.
provenance	Object	Server	Endpoint, input hash, timezone, settings and timestamp.
status	Enum	Server	pending, succeeded, failed or uncertain.

Contract guarantees

a. Plan enforcement and payment verification happen on the backend. Client-visible prices and counters cannot grant entitlement.
b. Use a request ID scoped to the user and operation. Reject the same ID with a different payload. Deduplicate payment events by provider event ID.
c. Store provider name, endpoint, calculation settings, input hash, schema version and calculation timestamp. Pin the plan and prompt version for each operation.
d. Provider calculations supply chart facts. AI output is an interpretation and must not override those facts. Payment webhooks and reconciliation establish paid access.

Implementation contracts will live in contracts/openapi.yaml and contracts/schemas/.

These are planned repository paths; freeze v1 schemas and representative fixtures during milestone M1.

7. Consistency, Idempotency, and Replay

Firestore transactions serialize entitlement changes and quota reservations. External provider execution cannot be made exactly once without vendor support. A durable operation record allows pending and uncertain attempts to be reconciled without charging the customer twice.

Scenario	Expected behavior	Reasoning
Duplicate request ID	Return saved output or pending state.	One customer quota debit per operation.
Reservation write fails	Return retryable error; no provider call.	Avoid untracked paid work.
Provider timeout	Mark uncertain; reconcile or resolve via support.	A timeout may still be billed by the vendor.
Plan changes mid-request	Use reserved plan version and period.	Prevent ambiguous quota changes.

8. Security and Privacy Considerations

a. Use Firebase Authentication with Google sign-in in the MVP. Firestore rules restrict users to their own profiles and saved results. Deny direct writes to payments, entitlements, usage and operations.
b. Collect only necessary birth and property information with clear consent. Do not place birth details, questions or access tokens in URLs, analytics or routine logs. Keep sensitive responses out of service-worker caches.
c. Store provider keys and webhook secrets in Secret Manager. Use separate development and production projects, least-privilege service identities and server-only credentials.
d. Require privileged roles for support tools. Keep provider calls and quota mutations behind authenticated routes; add per-user limits and App Check where supported. Disable sensitive debug logging in production.

e. Offer deletion of profiles and saved conversations. Propose 90-day chat retention and profile retention until deletion, subject to client approval. Keep legally required payment records separately; verify provider retention and region availability before launch.

9. Operational Readiness

Signal	SLO or alert	Owner	Launch gate
Successful requests	Proposed target >= 98% over 7 days	Developer	Pilot validation
Latency	Proposed p95: chart < 10s, chat < 30s	Developer	Measure vendor
Cost and wallet	Alert at 70% and 90% of budget	Client + dev	Required
Entitlement integrity	Zero duplicate grants in replay tests	Developer	Required
Recovery checks	Test timeout, rollback and account deletion	Developer	Required

Rollout: invite-only pilot; feature flags for paid calls and Prashna. Disable new paid work when the configured server budget is reached. Keep saved results readable.

10. Alternatives Considered

Alternative	Why considered	Decision
Native Android	Play Store distribution and native features	Defer; PWA minimizes initial build scope.
Direct browser API calls	Fewer backend components	Reject; exposes keys and bypasses quotas.
Own OpenAI or Gemini layer	Control prompts and response quality	Fallback if hosted chat fails validation; re-cost first.
Fixed AstrologyAPI plan	Predictable bill at higher volume	Begin with pay-as-you-go; assess eligible call mix.

11. Open Questions

a. Vendor validation: confirm Marathi quality, chat context and calculation provenance, Vastu response structure, commercial caching rights and failure billing.

b. Prashna: consultant must approve the question-time method, required coordinates, chart settings and interpretation workflow before this feature is enabled.

c. Commercial terms: confirm tax treatment, refunds, renewal terms, usage expiry, third-party invoices and actual launch traffic. Proposed prices are not an approved tariff.

d. Launch ownership: client supplies final logo, domain choice, privacy copy and consultant sign-off. Select Firebase region after checking function and database availability.

12. Decision and Next Steps

Proceed with a five-week, milestone-based PWA build after a vendor spike. Start with a small invited pilot, validate Marathi responses and payment reconciliation, then expand only after cost and error metrics meet the agreed gates.

Milestone	Deliverable	Exit criteria
M1 Week 1	Vendor spike and signed scope	Language, endpoints and cost assumptions validated
M2 Weeks 2 to 3	PWA, identity, chart, chat and Vastu	Owned data rules and core flows pass tests
M3 Week 4	Plans, checkout and usage ledger	Webhook replay and concurrency tests pass
M4 Week 5	Pilot, QA and client handover	Consultant sign-off; alerts and rollback demonstrated

13. UI Design and Customer Journeys

Use a warm ivory background, deep maroon primary actions and restrained saffron accents. A geometric sun mark is a proposed placeholder until the client supplies a logo. Use system Devanagari fonts and generous line height; keep all inputs and touch targets at least 44 pixels high.

Screen	Primary action	Required states
Home	Ask AI Guruji or create a birth chart	Marathi welcome, language control, usage summary
Birth Chart	Confirm birth data and generate	Missing or unknown time, place selection, loading, unavailable
AI Guruji	Ask and follow up	Profile needed, thinking, sourced response, quota exhausted
Vastu	Submit directions and room details	Missing direction, traditional guidance, practical advice
Prashna	Confirm question time and place	Method not enabled, confirmation, calculation failure
Plans	Choose a 30-day prepaid plan	Current plan, pending payment, confirmed payment
Consultation	Open WhatsApp	Consultant profile; separate human consultation fee

Navigation: Home, Guruji, Birth Chart, Vastu and More. More contains Prashna, Plans and Contact. On desktop, show these sections in a persistent sidebar. A language switch preserves the current flow and local input values; translate app copy with react-i18next and Ant Design component labels with ConfigProvider.

The prototype is a local interactive design, not a live service. It demonstrates all seven screens in Marathi, Hindi and English. Forms show a safe provider-unavailable state; no astrological results, payments or authentication are simulated as successful.

Detailed output layout: verified calculation summary, traditional interpretation, practical considerations, limits and consultation action. Display provider and calculation timestamp. Never show fabricated chart geometry or planetary values while loading. Unknown birth time blocks ascendant and time-sensitive conclusions.

Accessibility: visible focus, semantic labels, readable contrast, status announcements, keyboard access and error text next to the affected field. Respect reduced motion. Test at 360 px, 390 px and desktop widths with long Marathi labels.

14. Backend Interfaces and Data Model

Interface	Purpose	Controls
POST /v1/profiles	Save birth data and confirmed location	Ownership, consent, timezone validation
POST /v1/charts	Create or reuse a verified chart	Chart allowance; complete cache key
POST /v1/chat	Submit question or follow-up	AI allowance; session ownership; context limit
POST /v1/vastu	Basic directional assessment	Vastu allowance; validated directions
POST /v1/prashna	Question-time chart workflow	Feature flag; consultant-approved method
POST /v1/checkout	Create provider payment order	Server price and plan version
POST /v1/webhooks/razorpay	Verify and reconcile payment events	Raw-body signature; deduplication
GET /v1/me/usage	Show current period and allowances	Authenticated user only

Firestore collections: users/{uid}, profiles/{id}, conversations/{id}/messages/{id}, results/{id}, entitlements/{uid}, usage/{uid_period}, operations/{uid_requestId}, payments/{paymentId}, webhookEvents/{eventId}, and versioned plans/{planId}. Store ownerUid on owned records; use server timestamps and validate collection-specific rules.

Birth profile: local date and exact time, selected place label, latitude, longitude, IANA timezone, UTC instant and time certainty. Cache keys include normalized inputs, ayanamsa, chart settings, provider and calculation version. Natal data can be reused; transits and Panchang require date-sensitive keys and expiry.

AstrologyAPI adapter: geo_details, timezone_with_dst, astro_details, planets, horo_chart/:chart_id, major_vdasha and current_vdasha form the proposed chart bundle. Guruji uses hosted /api/chat with Standard or Advanced level. Basic Vastu uses score, findings and remedies. Confirm exact routes, auth, response shapes and language support in M1.

Payment flow: create order, complete checkout, show pending, receive signed webhook, verify captured payment and matching amount/currency, then activate a 30-day period atomically. Do not activate from a browser success callback. For MVP, prohibit overlapping purchases; refunds and disputes require recorded entitlement adjustments.

15. Commercial Model and Sources

Planning estimates carried forward from the pricing discussion on 17 October 2026. Validate the live vendor catalogue before purchase. AstrologyAPI hosted chat is the AI layer; a separate OpenAI or Gemini invoice is not included. Prices exclude applicable taxes and acquisition costs.

Plan and price	30-day allowance	Contribution at full use
Free INR 0	Lifetime: 3 AI questions and 1 chart	Cost INR 2.16 per activated trial
Basic INR 49	20 Standard AI, 1 chart, 1 Vastu	INR 36.12 or 73.7%
Standard INR 149	75 Standard AI, 3 charts, 3 Vastu	INR 102.80 or 69.0%
Premium INR 349	150 Advanced AI, 5 charts, 5 Vastu	INR 211.48 or 60.6%

Assumptions: chart bundle INR 0.66; Standard chat INR 0.50 per question; Advanced INR 0.80; basic Vastu bundle INR 0.75; payment reserve 3%. Each follow-up counts. Cached viewing is free; changed inputs count as a new calculation. No rollover. Contribution excludes fixed maintenance and build recovery.

Monthly scenario: 50 Basic, 35 Standard, 15 Premium and 100 new Free users. Revenue INR 12,900; API budget INR 4,200–4,700; Firebase INR 0–500; domain annualized INR 100–150; maintenance INR 1,500–2,500; payment reserve INR 387. Operating surplus INR 4,663–6,713 before taxes, marketing and development recovery. This is not guaranteed profit.

Developer proposal: INR 45,000 one time, split INR 18,000 / 18,000 / 9,000, with 30 days of defect fixes. Then INR 1,500–2,500 monthly for agreed support hours; new features quoted separately. Vendor bills belong to the client. A sub-INR 10,000 budget requires a smaller unpaid pilot.

Firebase uses Blaze billing for backend functions. Small-pilot estimates: Firestore INR 0–150; Hosting INR 0–100; Functions, secrets and logs INR 0–250. Google sign-in avoids SMS charges. Domain registration is separate and the proposed domain has not been checked for availability.

Sources: astrologyapi.com/credits-calculator; astrologyapi.com/developers/v1/chat-api; astrologyapi.com/developers/v1/guides/pay-as-you-go; firebase.google.com/pricing; razorpay.com/pricing. Hosted chat rates may be introductory. Marathi quality, Prashna compatibility and any additional endpoint costs remain launch gates.